A PRIME-POWER CONGRUENCE FOR AN ODD-EXPONENT BINOMIAL FAMILY

Abstract. For a nonnegative integer m, define

$$u_m(n) = \mathrm{Num}\left(\sum_{k=1}^{n} \frac{1}{k^{2m+1}} \binom{n}{k}^2 \binom{n+k}{k}^2\right),$$

where the numerator is taken after reducing the rational number to lowest terms. The conjecture treated here originates in the record for OEIS A357513, which states this parameterized form. We determine exactly when p^4 divides $u_m(p-1)$. For every prime p,

$$p^4 \mid u_m(p-1) \iff p-1 \nmid 2m+4 \text{ or } p \mid 2m+7.$$

Consequently, for each fixed m there are only finitely many exceptional primes; every such prime is at most $2m+5$.

1. Statement

The summation begins at $k = 1$. A version beginning at $k = 0$ is not defined because its first summand contains $1/0^{2m+1}$.

For a fixed nonnegative integer m, put

$$\mathcal{E}_m = \{p \text{ prime} : p-1 \mid 2m+4 \text{ and } p \nmid 2m+7\}.$$

Theorem 1. *Let $m \geq 0$ be an integer and let p be prime. Then*

$$p^4 \mid u_m(p-1) \iff p \notin \mathcal{E}_m.$$

Moreover, $\mathcal{E}_m$ is finite and is contained in the primes at most $2m+5$. If $p \in \mathcal{E}_m$, then

$$v_p\big(u_m(p-1)\big) = \begin{cases} 2, & p = 2, \\ 3, & p \text{ is odd.} \end{cases}$$

2. Common denominators

Fix m, and write

$$d = 2m+1.$$

For a prime p, let

$$S_{m,p} = \sum_{k=1}^{p-1} \frac{1}{k^d} \binom{p-1}{k}^2 \binom{p-1+k}{k}^2.$$

When p is odd, we work in the local ring

$$\mathbb{Z}_{(p)} = \left\{\frac{a}{b} : a, b \in \mathbb{Z},\ p \nmid b\right\}.$$

The reduced denominator of $S_{m,p}$ is coprime to p. Indeed, set

$$D_m = ((p-1)!)^d$$

and

$$N_m = \sum_{k=1}^{p-1} \left(\binom{p-1}{k}\binom{p-1+k}{k}\right)^2 \frac{D_m}{k^d}.$$

Since $k \mid (p-1)!$ for $1 \le k \le p-1$, every D_m/k^d is an integer. Thus $N_m \in \mathbb{Z}$, $S_{m,p} = N_m/D_m$, and $p \nmid D_m$.

Suppose that $S_{m,p} = A/B$ in lowest terms. Cross-multiplication gives $AD_m = N_m B$. Since $\gcd(A,B) = 1$, Euclid's lemma gives $B \mid D_m$. In particular, $p \nmid B$. Thus both B and D_m are p-adic units, so for every $r \ge 0$,

$$p^r \mid A \iff S_{m,p} \equiv 0 \pmod{p^r}. \tag{1}$$

In particular, a congruence for $S_{m,p}$ transfers directly to the reduced numerator $u_m(p-1)$.

3. The modular reduction

For $1 \le k \le p-1$, the exact identities

$$\binom{p-1}{k} = (-1)^k \prod_{j=1}^{k} \left(1 - \frac{p}{j}\right),$$

$$\binom{p-1+k}{k} = \frac{p}{k} \prod_{j=1}^{k-1} \left(1 + \frac{p}{j}\right)$$

give

$$\binom{p-1}{k}\binom{p-1+k}{k} = (-1)^k \frac{p}{k}\left(1 - \frac{p}{k}\right) \prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right). \tag{2}$$

After squaring (2) and dividing by k^d, the kth summand of $S_{m,p}$ is

$$\frac{p^2}{k^{d+2}} \left(1 - \frac{p}{k}\right)^2 \prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right)^2.$$

All denominators are units in $\mathbb{Z}_{(p)}$, and

$$\prod_{j=1}^{k-1} \left(1 - \frac{p^2}{j^2}\right)^2 \equiv 1 \pmod{p^2}.$$

The displayed summand already contains p^2. Hence, modulo p^4, it is congruent to

$$\frac{p^2}{k^{d+2}} - \frac{2p^3}{k^{d+3}}.$$

Put

$$r = d + 2 = 2m + 3, \qquad e = d + 3 = 2m + 4$$

and define

$$H_s = \sum_{k=1}^{p-1} \frac{1}{k^s} \in \mathbb{Z}_{(p)}.$$

Summing the preceding congruence gives

$$S_{m,p} \equiv p^2 H_r - 2p^3 H_e \pmod{p^4}. \tag{3}$$

The exponent r is odd. Since $k \mapsto p-k$ permutes $1, \dots, p-1$,

$$\begin{aligned} \frac{1}{(p-k)^r} &= -\frac{1}{k^r}\left(1 - \frac{p}{k}\right)^{-r} \\ &\equiv -\frac{1}{k^r} - \frac{rp}{k^{r+1}} \pmod{p^2}. \end{aligned}$$

Because $r+1=e$, summing over k yields

$$2H_r + rpH_e \equiv 0 \pmod{p^2}. \tag{4}$$

Multiply (4) by p^2 and double (3). Since $r+4=2m+7$, we obtain

$$2S_{m,p} \equiv -(2m+7)p^3 H_e \pmod{p^4}. \tag{5}$$

4. The finite-field obstruction

Reduction modulo p and the fact that inversion permutes $\mathbb{F}_p^\times$ give

$$H_e \equiv \sum_{x\in\mathbb{F}_p^\times} x^{-e} = \sum_{x\in\mathbb{F}_p^\times} x^e \pmod{p}.$$

The standard power-sum evaluation in the cyclic group $\mathbb{F}_p^\times$ is

$$H_e \equiv \begin{cases} 0 \pmod{p}, & p-1 \nmid e, \\ -1 \pmod{p}, & p-1 \mid e. \end{cases} \tag{6}$$

Assume first that p is odd. Then 2 is a unit in $\mathbb{Z}_{(p)}$. If $p-1\nmid e$, equations (5) and (6) give $S_{m,p} \equiv 0 \pmod{p^4}$. If $p-1 \mid e$, those equations instead give

$$2S_{m,p} \equiv (2m+7)p^3 \pmod{p^4}.$$

It follows that

$$S_{m,p} \equiv 0 \pmod{p^4} \iff p \mid 2m+7.$$

Together with (1), this proves the asserted classification for every odd prime.

For $p=2$, the sum has one term:

$$S_{m,2} = \binom{1}{1}^2 \binom{2}{1}^2 = 4.$$

Thus $u_m(1)=4$, so $2^4 \nmid u_m(1)$ and $v_2(u_m(1))=2$. This agrees with the criterion because $2-1 \mid 2m+4$, whereas $2 \nmid 2m+7$.

It remains to record the valuation at an odd exceptional prime. Equation (4) shows that $p \mid H_r$ in $\mathbb{Z}_{(p)}$. Equation (3) therefore gives $p^3 \mid S_{m,p}$. On the other hand, if $p-1 \mid 2m+4$ and $p \nmid 2m+7$, then (5) and (6) show that $S_{m,p} \not\equiv 0 \pmod{p^4}$. Consequently

$$v_p\big(u_m(p-1)\big) = 3.$$

Finally, if $p \in \mathcal{E}_m$, then $p-1$ is a positive divisor of $2m+4$. Therefore

$$p-1 \le 2m+4, \qquad p \le 2m+5.$$

This proves the finiteness assertion and completes the proof of Theorem 1.